Conference Date: 19.04.2024 **Referring:** Dr. Matteo Ruiz (Primary Care)

Patient: Eugene Price (26.08.57)

Primary diagnosis: Acute Myeloid Leukemia, **ELN 2022 Intermediate**, molecular **wild type** for NPM1/FLT3/IDH; normal karyotype at diagnosis
First-line therapy: 7+3 induction **started 01.11.2021** with cytarabine (7 days) + daunorubicin (3 days), followed by consolidation

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1) Initial Course (2021–2022)

- **Presentation (25.10.2021):** Fatigue, gingival bleeding, and easy bruising. **WBC 15.8×10⁹/L** (blasts 31%), **Hgb 7.9 g/dL**, **Plt 54×10⁹/L**, **LDH 612 U/L**. Coagulation normal.
- **Bone marrow:** 62% blasts; flow **CD34⁺/CD117⁺/HLA-DR⁺/CD13⁺/CD33⁺**.
- **Cytogenetics/Molecular:** 46,XY; myeloid NGS panel without NPM1/FLT3/IDH or other targetable mutations; DNMT3A VAF 14% noted.
- **Induction (01–07.11.2021):** Expected pancytopenia; one **febrile neutropenia** episode, culture-negative.
- **Day-28 marrow (29.11.2021): CR** (blasts <2%), MRD by flow **<0.1%**; DNMT3A persisted as clonal hematopoiesis (not the AML clone).
- **Consolidation: IDAC ×3 cycles** (Dec 2021–Feb 2022). Toxicities: grade 2 mucositis, transient transaminitis; no neurotoxicity.
- **Surveillance 2022:** Counts stable; port removed July 2022. No maintenance given intermediate risk and recovery.

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2) Relapse/Progression (Spring 2023)

- **Early signal:** March–April 2023 with **falling platelets (to 78×10⁹/L)** and rising LDH (to 480 U/L).
- **Bone marrow 01.05.2023: Relapsed AML** with **28–30% blasts**. Cytogenetics: new subclone with **+8**; NGS: **RUNX1** p.R201Q (VAF 9%) emerged; DNMT3A stable at 16%.
- **Performance status/Echo:** ECOG 1; LVEF 59%.
- **Comorbidities impacting choice:** Type 2 diabetes (A1c 7.0%), **CAD** s/p PCI (2015) on **aspirin** (held when Plt <50), **CKD 3a** (Cr 1.3–1.5), mild COPD.

Consensus: Given age/comorbidities but good performance status, aim for **remission re-induction with venetoclax + azacitidine** to bridge to **reduced-intensity allo-HSCT** if response adequate.

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3) Second-Line Therapy (May–Sept 2023)

- **Regimen: Azacitidine 75 mg/m² D1–7 q28d + Venetoclax** (ramp to 400 mg D1–21).
- **Prophylaxis:** Acyclovir 400 mg bid; **levofloxacin** during neutropenic nadirs; **mold-active azoles avoided** (CYP3A interaction) → used **micafungin** when ANC <0.2.
- **Toxicities:** Cycles 1–2 with grade 4 neutropenia and grade 3 thrombocytopenia; single admission for FN (cultures negative). No TLS.
- **Response:**
  - **Marrow 10.07.2023: CRi** (blasts 2–3%), MRD by flow 0.15%.
  - **Marrow 18.09.2023: CR** with **MRD 0.04%**; platelets improving (to 92×10^9/L), ANC ~1.0×10^9/L.
- **Decision:** Proceed with **HLA typing** and transplant evaluation.

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4) Transplant Work-up (Oct–Dec 2023)

- **Donor search:** Identified **10/10 matched unrelated donor** (O+, CMV+).
- **Pre-transplant evaluation:**
  - **Cardiac:** LVEF 55–57%; no inducible ischemia on stress echo; continue **metoprolol 50 mg qd**, **atorvastatin 40 mg qHS**; **aspirin 81 mg** only when Plt >50.
  - **Pulmonary:** FEV_1 75% predicted; DLCOcorr 70% (mild COPD).
  - **Renal:** eGFR 52–58 mL/min/1.73 m².
  - **Infectious serologies:** CMV IgG+, EBV IgG+; PCR negative. Vaccinations updated.
- **Bridging therapy: Azacitidine + venetoclax** continued with **venetoclax shortened to D1–14** due to cumulative neutropenia.
- **Chimerism planning/MRD: RUNX1** mutation became undetectable on peripheral blood NGS in December; flow MRD <0.02%.

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5) Current Status & Plan (Conference 19.04.2024)

Symptoms: Feels well, walking 20–30 minutes daily, no infectious symptoms or bleeding.
Exam: Afebrile; BP 124/70; HR 72; SpO_2 98% RA; no lymphadenopathy; lungs clear; abdomen benign; no petechiae. ECOG **1**.
Laboratory data (15.04.2024): WBC **4.1×10^9/L** (ANC **1.6**), Hgb **11.8 g/dL**, Plt **138×10^9/L**; Na 138, K 4.3, Cr **1.32**, AST/ALT **26/22**; **LDH 190**.
MRD: Flow **<0.02%** (PB); targeted NGS **negative** for **RUNX1**; **DNMT3A 14%** consistent with CH.
Imaging: CXR clear; echocardiogram LVEF 56% (04.2024).
Line: None (port removed 02.2022; PICC as needed during cycles).

Plan (agreed by Hematology, BMT, Cardiology, ID):

1. **Proceed to RIC allogeneic HSCT** late April/early May using matched unrelated donor. Conditioning **Fludarabine/Busulfan (AUC-targeted)** given renal function; **GVHD prophylaxis** with **post-transplant cyclophosphamide + tacrolimus + mycophenolate**.

2. **Disease control until admission:** Continue **azacitidine** (D1–5 “mini” cycle) with **venetoclax D1–7** only if ANC >0.7 and Plt >75; otherwise **hold** to allow count recovery before conditioning.
3. **Supportive care:**
   - Antimicrobials: continue **acyclovir**; start **levofloxacin** if ANC <0.5; begin **mold-active azole** post-conditioning (posaconazole with venetoclax held).
   - Transfusions: irradiated, leukoreduced, CMV-safe products; thresholds Hgb ≥8, Plt ≥10 (≥20 if bleeding).
   - Electrolytes: maintain K^+ >4.0, Mg^{2+} >2.0; monitor QTc if azoles needed.
4. **Cardio/renal optimization:** Metoprolol and ACE inhibitor (lisinopril 5 mg qd) to continue; avoid NSAIDs; hydrate pre-contrast.
5. **Survivorship & counseling:** Fertility not applicable; discuss infection risks, GVHD symptoms, vaccination schedule post-transplant.
6. **Contingency:** If counts fall or MRD rises prior to transplant, consider **gilteritinib is not applicable** (FLT3 wt); alternatives include **FLAG-lite** or **clinical trial**; however, preference is to proceed directly to transplant while in deep remission.

Bottom line: Mr. Price relapsed ~May 2023, responded to **venetoclax + azacitidine**, and is now in **morphologic CR with very low/undetectable MRD**, moving forward to planned **reduced-intensity allo-HSCT** as the best chance for durable control.

Attendees: Hematology (MD/PA), BMT physician, Cardiology, Infectious Diseases, Pharmacist, Social Work.
Dictated by: [Name], MD — Hematology